

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No.

B.E. II (ME)

3

ET

SECOND B.E. EXAMINATION - 2026

MECHANICAL ENGINEERING

(III SEMESTER CBCS)

**3ME41A (ME): ENGINEERING
THERMODYNAMICS**

Time - Three Hours

Maximum Marks - 70

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) State Zeroth law of thermodynamics. Derive the unsteady flow energy equation for filling a tank assuming the tank is initially empty and insulated.

7

J-3853

1

(Contd.)

- (b) State Kelvin Planck and Clausius statements of second law of thermodynamics. State and prove any one corollary of second law of thermodynamics.

7

2. (a) A one kg of water at 273K is brought into contact with a heat reservoir at 373K. When the water has reached 373K, find the entropy change of water, of the heat reservoir, and of the universe. (b) If the water is heated from 273K to 373K by first bringing it in contact with a reservoir at 323K and then with a reservoir at 373K, what will the entropy change of the universe be?

7

- (b) Derive the expression of exergy balance for an open system.

7

3. (a) A fluid at 200kPa and 300°C has a volume of 0.8m³. In a frictionless process at constant volume the pressure changes to 100kPa. Find the final temperature and the heat transferred if the fluid is steam.

7

Saturated Pressure P_{sat}	Saturated temperature T_{sat}	v_f	v_g	u_f	u_{fg}
50kPa	81.33°C	0.00103 m ³ /kg	3.24 m ³ /kg	340.42 kJ/kg	2143.4 kJ/kg
Superheated steam					
P_{sat}	T	V	u		
100 kPa (T_{sat} 99.62)	300°C	2.6388m ³ /kg	2810.4 KJ/kg		

J-3853

2

(Contd.)

- (b) State Dalton's law of partial pressure and Amagat's law of partial volumes. 7
4. (a) Explain combustion process, stoichiometric air fuel ratio and excess air. 7
- (b) Write combustion equation and find stoichiometric air fuel ratio for fuel C_6H_{14} . 7
5. (a) Derive four Maxwell's equations. 7
- (b) Derive the following Clausius - Clapeyron equation - 7

$$\frac{dP}{dT} = \frac{h_{fg}}{T(v_g - v_f)}$$

Write the use of the equation.

6. (a) Define and derive the expression of volumetric efficiency of a single stage compressor with clearance. Also discuss how the volumetric efficiency varies with clearance and pressure ratio. 7
- (b) Classify the air compressors. Write the purpose of inter-cooling and after-cooling in reciprocating air compressors. 7
7. Write short notes on any four - $4 \times 3.5 = 14$
- (i) Carnot cycle
- (ii) Joule-Kelvin expansion

- (iii) Second law efficiency
- (iv) Vander Waals gas equation
- (v) Clausius inequality
- (vi) Exergy and Anergy

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No.

B.E. II (ME)

3

MT

SECOND B.E. EXAMINATION - 2026

MECHANICAL ENGINEERING

(III SEMESTER CBCS)

3ME42A (ME): MATERIALS TECHNOLOGY

Time - Three Hours

Maximum Marks - 70

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Define different type of Packing arrangement with neat sketch and details. 7

- (b) Describe different types of defect in metals and alloys with detail information. 7
2. (a) What is the role of dislocation in metals & alloy? How its effect the different properties in the metals? 7
- (b) Define recovery and recrystallization & how its effected by the temperature. 7
3. (a) Discuss the Iron-Iron carbon equilibrium diagram with neat sketch and details. 10
- (b) Describe the Components and Phase with examples. 4
- 4 (a) Discuss the TTT diagram with detail information with examples. 10
- (b) What is Case hardenability & case hardening? 4
5. (a) Describe the non-metallic materials behaviour and how its improve the different mechanical properties. 7
- (b) What is the effect of Alloying elements in steels? Describe in details. 7

6. (a) Lead is having FCC structure with interatomic distance as $3.499 \text{ \AA}$. Find the number of atoms per sq. mm in (100) plane. 7
- (b) Write a short note on Sintering, Compaction and Metallic powders. 7
7. Write a short note on: 14
- (a) Cooling Curves
- (b) Hardening and Quenching
- (c) Smart materials
- (d) Different mech. properties.

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No. 250mmEC1046

B.E. II (ME)

3

KOM

SECOND B.E. EXAMINATION - 2026

MECHANICAL ENGINEERING

(III SEMESTER CBCS)

3ME43A(ME): KINEMATICS OF MACHINES

Time - Three Hours

Maximum Marks - 70

Note:- (i) Q. No. 1 is compulsory & complete solution (including calculations) for Q. No. 1 is to be given in the drawing sheet.

(ii) Attempt any four questions from question no. 2 to 7.

(iii) All questions carry equal marks.

(iv) Any missing data may be suitable assumed and stated clearly.

(v) Draw neat sketches, wherever necessary.

1. The dimensions of the various links of a pneumatic riveter as given in Figure 1 are as follows: $OA = 175$ mm; $AB = 180$ mm; $AD = 500$ mm; and $BC = 325$ mm. Find the velocity ratio between C and ram D when OB is vertical. What will be the efficiency of the machine if a load of 2.5 kN on the piston C causes a thrust of 4 kN at the ram D?

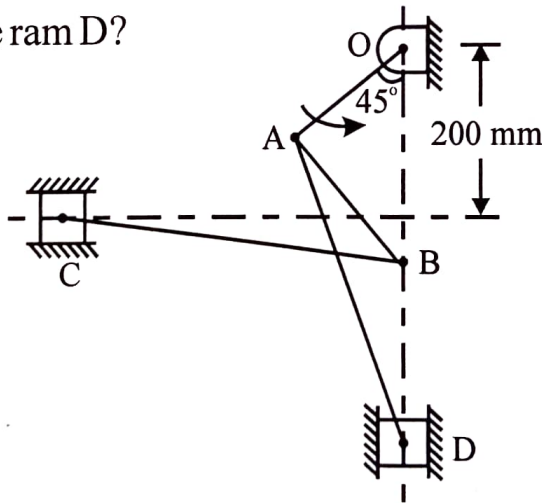


Figure 1.

2. (a) Explain Klein's construction theorem, with its applications.

7

- (b) State Arnold-Kennedy theorem of three centres.
How instantaneous centres are located? 7
3. (a) Discuss various types of constrained motions. 7
- (b) In a quick-return motion mechanism of the crank and slotted lever type, the ratio of maximum velocities is 2. If the length of stroke is 250 mm, find (a) the length of the slotted lever, (b) the ratio of times of cutting and return stroke, and (c) the maximum cutting velocity in m/sec if the crank rotates at 30 rpm. 7
4. (a) Derive an expression for the ratio of shafts velocities for Hook's joint and draw the polar diagram depicting the salient features of driven shaft speed. 7
- (b) Explain uniform pressure and uniform wear theories for the flat collar bearings. 7
5. (a) Deduce expression for the ratio of tight and slack side tension in case of a V-belt drive. 7
- (b) Determine the maximum power that can be transmitted through a flat belt having the following data: cross-section of belt = 300 mm x 12 mm; ratio of friction tensions = 2.7; maximum permissible tension in belt = 2.8 N/mm^2 ; mass density of the belt material = 0.0015 g/mm^3 . 7

6. (a) A screw jack is used to raise a load of 5 tonnes (1 tonne = 9.81 kN). The pitch of single-start square threads used for the screw is 24 mm. The mean diameter is 72 mm. Determine the force to be applied at the end of 1.2 m long handle when the load is lifted with constant velocity and rotate with spindle. Take $\mu = 0.2$. Also find the mechanical efficiency of the jack. 7
- (b) Explain the working of a multi-plate clutch with a help of neat-sketch. 7
7. Write short notes on any four:
- (a) Discuss Coriolis component of acceleration. When it occurs?
- (b) State the law of friction.
- (c) Explain the centrifugal tension in a belt drive and reduce expression with neat sketch.
- (d) Discuss the Kutzbach and Grubler's criterion for degree of freedom for planar mechanism.
- (e) Discuss the construction of belt and various types of belt materials. 4x3.5

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No.

B.E. II (ME)

3

Mech. Of Solids

SECOND B.E. EXAMINATION - 2026

MECHANICAL ENGINEERING

(III SEMESTER CBCS)

3ME44A: MECHANICS OF SOLIDS

Time - Three Hours

Maximum Marks - 70

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (A) A reinforced concrete circular section of 50,000 mm² cross-sectional area carries 6 reinforcing bars whose total area is 500 mm². Find the safe load, the column can carry, if the concrete is not to be

- stressed more than 3.5 MPa. Take modular ratio for steel and concrete as 18. 7
- (B) Obtain a relation for the elongation of a uniformly circular tapering section. 7
2. A simply supported beam of length 6 m carries point loads of 1, 2 and 3 kN at a distance of 1.5, 3 and 4.5 m respectively from left support. There is a udl of 1.5 kN/m over the entire length. Draw the SFD and BMD. 14
3. A horizontal steel girder having uniform cross-section is 14 m long and is simply supported at its ends. It carries 12 kN at 3 m and 8 kN at 9.5m from left support. Calculate the deflections of the beam under the two loads. Take $E = 200 \text{ GPa}$ and $I = 160 \times 10^6 \text{ mm}^4$. 14
4. A solid alloy shaft of 50 mm diameter is to be coupled in series with a hollow steel shaft of the same external diameter. Find the internal diameter of the steel shaft if the angle of twist per unit length is to be 75 percent of that of the alloy shaft. Determine the speed at which the shafts are to be driven to transmit 185 kW if the limits of shearing stress are to be 55 and 85 MPa in the alloy and steel respectively. $C_{\text{steel}} = 2.2 C_{\text{alloy}}$. 14

5. (A) Derive Torsion equation mentioning assumptions. 7
- (B) A cantilever beam is rectangular in section having 80 mm width and 120 mm depth. If the cantilever is subjected to a point load of 6 kN at the free end and the bending stress is not to exceed 40 MPa, find the span of the cantilever beam. 7
6. (A) At a point in a two dimensional system the normal stress on two mutually perpendicular planes are σ_1 and σ_2 and shear stress is τ . Show that one of the principal stresses is zero if $\tau = \sqrt{(\sigma_1 \times \sigma_2)}$. 7
- (B) The principal tensile stress at a point across two mutually perpendicular planes are 100 N/mm². and 50 N/mm² Determine the normal and tangential stresses on a plane inclined at 30° to the axis of minor principal stress. 7
7. (A) Explain the limitation of Euler's formula for columns. 7
- (B) Derive Torsion equation for shafts stating the assumptions. 7

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No.

B.E. II (Maths.)

14

Maths

SECOND B.E. EXAMINATION - 2026

MATHEMATICS

(III SEMESTER CBCS)

3ME21A (MA) : MATHEMATICS

Time - Three Hours

Maximum Marks - 70

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Solve the simultaneous differential equations

$$t dt - x dt = t dx + x dt ;$$

$$t dt + t dy = t(x+y) dt + 2x dt$$

J-3857

1

(Contd.)

- (b) Solve the total differential equation

$$(ay - bz) dx + ((cz - ax) dy + (bx - cy) dz = 0$$

2. (a) Solve the partial differential equation

$$4 \frac{\partial^2 u}{\partial x^2} + 12 \frac{\partial^2 u}{\partial x \partial y} + 9 \frac{\partial^2 u}{\partial y^2} = \cos(2y + 3x)$$

- (b) Find general solution of one dimensional wave equation, using separation of variables method.

3. (a) For the following data obtain the forward and backward difference polynomials and hence interpolate at $x = 0.25$ and $x = 0.35$

x	:	0.1	0.2	0.3	0.4	0.5
f(x)	:	1.40	1.56	1.76	2.00	2.28

- (b) For the following data estimate the value of $f(x)$ at $x = 0.25$, using Bessel's interpolation formula

x	:	0.0	0.1	0.2	0.3	0.4	0.5
f(x)	:	-1.5	-1.27	-0.98	-0.63	-0.22	0.25

4. (a) Using Simpson's 3/8 rule, find approximate value of the integral ($h = 0.1$)

$$\int_0^1 \frac{1}{\sqrt{x(1-x)}} dx$$

- (b) Perform four iterations of the Newton Raphson method to find smallest positive root of the equation $x^3 - 5x + 1 = 0$

5. (a) Using Jacobi's method solve the system of Linear equations

$$3x - y = 2; 2x - 3y + z = 0; 2y - 3z + 1 = 0$$

- (b) Using Euler's method, find approximate value of y for $x = 1.5$ with step size $h = 0.1$, from the differential equation $\frac{dy}{dx} = x^2 + y^2$; $y(1) = 2$.

6. (a) Find Laplace transform of the function

$$f(t) = \frac{1}{t\sqrt{t}} e^{-k^2/4t}; \quad k > 0$$

- (b) Find inverse Laplace transform of the function

$$f(s) = \frac{1}{(s^2 + 2s + 5)^{3/2}}$$

7. (a) Using Laplace transform, solve the initial value problem

$$\frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} + 4y = \sin wx; \quad x > 0$$

$$\text{and } y(0) = y_0, y'(0) = y_1$$

- (b) Find $f(x)$, if its Fourier Sine transform is $\frac{P}{(1 + P^2)}$

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No.

B.E. II (ME)

3

PMT

SECOND B.E. EXAMINATION - 2026
MECHANICAL ENGINEERING
(III SEMESTER CBCS)
3ME45A (ME): PRODUCTION MACHINE
TOOLS

Time - Three Hours

Maximum Marks - 70

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) What are the types of lathes you are familiar with and give their applications? Explain the procedure for turning a job, which is very long. List the various operations and tools (cutting and holding) required for the operation. 7

- (b) Draw a setup required for cutting external threads on a centre lathe. Explain clearly the various methods that are followed for the purpose of engaging the cutting tool for giving the depth of cut in threading. 7
2. (a) Describe with a simple sketch the operation of a Swiss type (sliding headstock) automatic lathe in terms of operations possible, tools used, method of setting and types of components made on the lathe. 7
- (b) Briefly explain with neat sketches the types of work holding devices that are commonly employed in automatic lathes. Specify the limitations of them. 7
3. (a) What is a planing machine? Explain the principal parts of a planing machine with neat sketches. 7
- (b) Describe with a schematic sketch the operation of a hydraulic shaper. Give the advantages of Hydraulic shaper compared to mechanical (crank) shaper. 7

- 4 (a) Differentiate between up-milling and down milling. Explain their applications mentioning the most commonly used method. 7
- (b) Explain the construction of a dividing head giving the applications for which it can be used. Calculate the indexing requirement for 127 divisions on a milling machine equipped with a differential indexing head. The index plates available are:
- Plate no. 1: 15, 16, 17, 18, 19, 20 holes
- Plate no. 2: 21, 23, 27, 29, 31, 33 holes
- Plate no. 3: 37, 39, 41, 43, 47, 49 holes
- The change gear set available is : 24, 24, 28, 32, 40, 44, 48, 56, 64, 72, 86, 100 7
5. (a) Explain briefly the construction of a radial drilling machine with emphasis on how the requisite motions are obtained. A hole of 40 mm diameter and 50 mm deep is to be drilled in mild steel component. The cutting speed can be taken as 65 m/min and the feed rate as 0.25 mm/rev. Calculate the machining time and the material removal rate.

7

- (b) How is grinding different from other machining operations? Explain its applications in view of its capabilities. Compare grinding, honing and lapping operations. 7
6. (a) Describe the principle of spot welding with the help of a neat diagram. 7
- (b) What are the different types of jig bushes used? Discuss the different parts that should form part of a milling fixture. 7
7. (a) What do you understand by the word "canned cycle" in manual part programming? Explain with neat sketches the differences between the operation of the canned cycles G81 and G83. 7
- (b) Explain the concept and need for a post processor as used in computer assisted part programming systems such as APT. Describe the functions of a post processor. 7